

This document contains set of questions for a game development, each being an enhancement for the game.

Zip the program as you solve each question and upload them separately in the answers submission form.

1. Write a program to fill up the matrix size of 9*9 with 0 . Generate random number for position **x** and position **y** and place '**5**' in that generated (x,y) position. Here, **5** is gun.

Note : Random number should be generated between 1 to 9. Assume matrix position starts as (1,1)

Input :

0 - Filled matrix :

[illegible]

Generated random number for 'x' : 9

Generated random number for 'y' : 1

Output :

Gun placed matrix :

[illegible]

5	0	0	0	0	0	0	0	0
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2. For the previous program, now add player '1', position him in the location by generating 'x' and 'y' by random number.

Note : random number generated for gun position (x,y) and player position (x,y) should not be same. In case of generating same values, regenerate for new position. Player 1 denote as '1'

0 - Filled matrix :

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Placing Gun :

Generated random number for 'x' : 3

Generated random number for 'y' : 4

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Placing player '1' :

Generated random number for 'x' : 3

Generated random number for 'y' : 4
as already Gun (5) placed in same location. Generate new position values for player '1' :

Generated random number for 'x': 9

Generated random number for 'y' : 1

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0

3. **Moving operation** : Now introduce moving functionality to the player. Finish the game when reaching gun [5] position

- 1 - Choosing left direction
- 2 - Choosing right direction
- 3 - Choosing up direction
- 4 - Choosing down direction.

For each move generate number from 0-3 to move forward number of position. Forward 0 means no move.

0 - Filled matrix :

[illegible]

Placing Gun :

Generated random number for 'x': 3

Generated random number for 'y' : 4

[illegible]

Placing player '1' :

Generated random number for 'x': 9

Generated random number for 'y' : 1

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0

Enter the direction you want to move : 2

Number of move (random number: 0-3) : 1

[illegible]

[illegible]

Game Over !!!

4. **Two Player game** : Now introduce player '2' . Player who gets gun (reaching gun position) is winner

Note : Generated position of player '2' should not be same as player '1' and gun [5] position.

0 - Filled matrix :

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Placing Gun :

Generated random number for 'x' : 3

Generated random number for 'y' : 4

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Placing player '1' :

Generated random number for 'x' : 9

0	0	0	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---

Gun Holder : -

Player '2' move :

Enter the direction you want to move : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Player 2 Picked Gun - So He is the winner !!!

5. Add below functionality. Player who gets gun can shoot other person when facing/meet opponent in the same row or column.

- i) Player who gets gun should kill opponent in 3 tries (moves) , if so he is the winner
- ii) If player who gets gun was not able to shoot the opponent in 3 tries (moves), then opponent is the winner.
- iii) If player who gets gun is in same row/column in the opponent , before move shooting is possible. For shooting , enter option '**5**'

Test case : follow the previous program (**4**) test case. So, now player '**2**' gets gun.

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '1' move :

Enter the direction you want to move : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '2' move :

Enter the direction/shooting option : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '1' move :

Enter the direction/shooting option : 2

Number of move (random number: 0-3) : 3

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '2' move :

Enter the direction/shooting option : 5

Player 2 shoot the opponent - So He is the winner !!!

6. Now, Instead of finishing game, in case of missing opponent to shoot. If the player who gets gun was not able to shoot the opponent, then gun is freed from the player and will be placed randomly in some other places [x,y - randomly generated position]. Now, both the player having chance to pick the Gun and game will be continued till some one gets shot.

Condition : Freed gun should be placed in new position which is within the radius of distance 2 from the Player who is holding gun. Ex. Assume Player '1' holding gun at position (3,4) and his tries are over. Now, Gun could be placed in following positions:

(1,2), (1,3), (1,4), (1,5) , (1,6)
 (2,2), (2,3), (2,4), (2,5) , (2,6)
 (3,2), (3,3), , (3,5), (3,6)
 (4,2), (4,3), (4,4), (4,5) , (4,6)
 (5,2), (5,3), (5,4), (5,5) , (5,6)

Test case : follow the program - **4** test case. So, now player '**2**' gets gun.

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '1' move :

Enter the direction you want to move : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '2' move :

Enter the direction/shooting option : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player '1' move :

Gun Holder : Player 1

Player '1' move :

Enter the direction/shooting option : 1

Number of move (random number: 0-3) : 1

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 1

Player '2' move :

Enter the direction/shooting option : 1

Number of move (random number: 0-3) : 0

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Player '1' move :

Enter the direction/shooting option : 5

Player 1 shoot the opponent - So He is the winner !!!

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	6

Placing player '1' :

Generated random number for 'x' : 7

Generated random number for 'y' : 4

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	6	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Placing player '2' :

Generated random number for 'x' : 1

Generated random number for 'y' : 4

0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	6	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : -

Player 1' Survival Power : 0

Player 2 Survival Power : 0

Player '1' move :

Enter the direction/shooting option : 3

Number of move (random number: 0-3) : 1

0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	5	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : -

Player 1' Survival Power : 1

Player 2 Survival Power : 0

Player '2' move :

Enter the direction/shooting option : 4

Number of move (random number: 0-3) : 2

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Gun Holder : Player 2

Player 1' Survival Power : 1

Player 2 Survival Power : 0

Player '1' move :

Enter the direction/shooting option : 4

Number of move (random number: 0-3) : 0

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	2	0	0	0	0	0

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

and so on....